

CENTRALIZED WEB-BASED HEALTH DOCUMENT AND APPOINTMENT MANAGEMENT SYSTEM

A Minor Project Report

Submitted in partial fulfillment of requirement of the

Degree of

**BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE &
ENGINEERING**

BY

Yashasvi Panchal (Enrollment No. EN23CS3011158)

Yashika Mehta (Enrollment No. EN23CS3011159)

Yashwini Shinde (Enrollment No. EN23CS3011164)

Under the Guidance of

Prof. Priyanka Jain (Internal Guide)

Dr. Hare Ram Jha (Internal Guide)



Department of Computer Science & Engineering

Faculty of Engineering

MEDICAPS UNIVERSITY, INDORE- 453331

APRIL-2026

CENTRALIZED WEB-BASED HEALTH DOCUMENT AND APPOINTMENT MANAGEMENT SYSTEM

A Minor Project Report

Submitted in partial fulfillment of requirement of the

Degree of

**BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE &
ENGINEERING**

BY

Yashasvi Panchal (Enrollment No. EN23CS3011158)

Yashika Mehta (Enrollment No. EN23CS3011159)

Yashwini Shinde (Enrollment No. EN23CS3011164)

Under the Guidance of

Prof. Priyanka Jain (Internal Guide)

Dr. Hare Ram Jha (Internal Guide)



Department of Computer Science & Engineering

Faculty of Engineering

MEDICAPS UNIVERSITY, INDORE- 453331

APRIL-2026

Report Approval

The project work "**Centralized Web-based Health Document and Appointment Management System**" is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

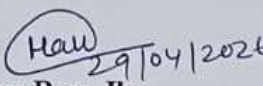
It is to be understood that by this approval the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion drawn there in; but approve the "Project Report" only for the purpose for which it has been submitted.

Internal Examiner

Prof. Priyanka Jain

Department of Computer Science & Engineering

Medicaps University, Indore


Dr. Hare Ram Jha

Department of Computer Science & Engineering

Medicaps University, Indore

Certificate

We, Prof. Priyanka Jain and Dr. Hare Ram Jha certify that the project entitled “Centralized Web-based Health Document and Appointment Management System” submitted in partial fulfillment for the award of the degree of Bachelor of Technology/Master of Computer Applications by Yashasvi Panchal, Yashika Mehta, Yashwini Shinde is the record carried out by him/them under my/our guidance and that the work has not formed the basis of award of any other degree elsewhere.

Prof. Priyanka Jain

Department of Computer Science &
Engineering

Medicaps University, Indore

Dr. Hare Ram Jha

Department of Computer Science &
Engineering

Medicaps University, Indore

Dr. Kailash Chandra Bandhu

Head of the Department
Computer Science & Engineering
Medicaps University, Indore

Declaration

I/We hereby declare that the project entitled “**Centralized Web-based Health Document and Appointment Management System**” submitted in partial fulfillment for the award of the degree of Bachelor of Technology in ‘Computer Science & Engineering’ completed under the supervision of **Prof. Priyanka Jain and Dr. Hare Ram Jha**, Department of Computer Science and Engineering, Medicaps University, Indore is an authentic work.

Further, we declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

Signature and name of the student(s) with date

1. Yashasvi Panchal (EN23CS3011158) Yashasvi Date: 29/04/26
2. Yashika Mehta (EN23CS3011159) Yashika Date: 29/04/26
3. Yashwini Shinde (EN23CS3011164) Yashwini Date: 29/04/26

Acknowledgements

I would like to express my deepest gratitude to the Honorable Chancellor, **Shri R C Mittal**, who has provided me with every facility to successfully carry out this project, and my profound indebtedness to **Prof. (Dr.) D. K. Patnaik**, Vice Chancellor, Medicaps University, whose unfailing support and enthusiasm has always boosted up my morale. I also thank **Dr. Ratnesh Litoriya**, Dean, Faculty of Engineering, Medicaps University, for giving me a chance to work on this project. I would also like to thank my Head of the Department **Dr. Kailash Chandra Bandhu** for his continuous encouragement for the betterment of the project.

We extend our heartfelt thanks to our Internal Guide, **Prof. Priyanka Jain**, and **Dr. Hare Ram Jha**, for their invaluable guidance, constant support, and constructive feedback throughout the development of SimplifyRepo. Their expertise in software engineering and AI-enabled systems helped us navigate complex technical decisions with confidence.

We also acknowledge the open-source communities behind React.js, Node.js, Express.js, Mermaid.js, and OpenAI's API documentation, which provided indispensable references during implementation

It is their help and support, due to which we became able to complete the design and technical report.

Without their support this report would not have been possible.

Yashasvi Panchal, Yashika Mehta, Yashwini Shinde

B.Tech. III Year (T)

Department of Computer Science & Engineering

Faculty of Engineering

Medicaps University, Indore

Abstract

The Centralized Web-Based Health Document and Appointment Management System is a comprehensive digital platform designed to simplify the management of healthcare records and appointment scheduling for individuals and families. In the modern healthcare environment, patients and their families frequently struggle with disorganized paper records, fragmented medical histories across multiple providers, and inefficient appointment booking processes. This project addresses these challenges by providing a secure, web-based solution that centralizes the storage and management of health documents while integrating an appointment reminder system.

The system supports multiple user profiles including parent and child accounts, allowing a single user to manage health records for entire families. Core functionalities include user registration and authentication, document upload and management with categorization, appointment scheduling with automated reminder notifications, medication tracking, and secure role-based access control. The platform is built using modern web technologies including React.js for the frontend, Node.js and Express.js for the backend, and MongoDB for the database.

The proposed system aims to reduce administrative burden on patients, improve continuity of healthcare, and enhance patient participation in health management. By digitizing and centralizing health records, the system eliminates the risk of document loss and enables authorized healthcare providers to access complete patient histories during consultations. The appointment reminder feature helps users adhere to their healthcare schedules, reducing missed appointments and improving health outcomes.

This report presents the complete development lifecycle of the system, including requirements analysis, system design, implementation details, and testing strategies. The design incorporates Data Flow Diagrams, Entity Relationship Diagrams, and Use Case Diagrams to document system behavior. The outcome is a scalable, secure, and user-friendly platform that contributes to the digital transformation of personal health management.

Keywords: Health Document Management, Appointment Reminder System, Web-Based Healthcare Platform, Personal Health Records, Family Profile Management, React.js, Node.js, MongoDB, Role-Based Access Control, Digital Health, Medication Tracking.

Table of Contents

		Page No.
	Report Approval	ii
	Declaration	iii
	Certificate	iv
	Acknowledgement	v
	Abstract	vi
	Table of Contents	vii
	List of figures	ix
	List of tables	x
	Abbreviations	xi
	Notations & Symbols	xii
Chapter 1	Introduction (Whichever is applicable)	
	1.1 Introduction	1
	1.2 Literature Review	
	1.3 Objectives	
	1.4 Significance	
	1.5 Research Design	
	1.6 Source of Data	
	1.7 Chapter Scheme	
Chapter 2	REQUIREMENTS SPECIFICATION	
	2.1 User Characteris	
	2.2 Functional Requirements	
	2.3 Dependencies	
	2.4 Performance Requirements	
	2.5 Hardware Requirements	
	2.6 Constraints & Assumptions	
Chapter 3	DESIGN (Whichever is applicable)	
	3.1 Algorithm (if Applicable)	
	3.2 Function Oriented Design for procedural approach	
	3.3 System Design (Whichever is applicable)	
	3.3.1 Data Flow Diagrams (Level 0,Level1)	

	3.3.2 Activity Diagram	
	3.3.3 Flow Chart	
	3.3.4 Class Diagram	
	3.3.5 ER Diagram	
	3.3.6 Sequence diagram	
	3.4 Database Design	
	3.4.1 Logical Database Design	
	3.4.2 Physical Database Design	
Chapter 4	Implementation, Testing, and Maintenance	
	4.1 Introduction to Languages, IDE's, Tools and Technologies used for Implementation	
	4.2 Testing Techniques and Test Plans (According to project)	
	4.3 Installation Instructions	
	4.4 End User Instructions	
Chapter 5	Results and Discussions	
	5.1 User Interface Representation (of Respective Project)	
	5.2 Brief Description of Various Modules of the system	
	5.3 Snapshots of system with brief detail of each	
	5.4 Back Ends Representation (Database to be used)	
	5.5 Snapshots of Database Tables with brief description	
Chapter 6	Summary and Conclusions	
Chapter 7	Future scope	
	Appendix	
	Bibliography	
	List of Publications (If any)	
	Reprints of publications(If any)	
Note	Results and Discussions may not be a separate chapter it may be included in main chapter	

List of Figures

Figure No.	Figure Title	Chapter
3.1	System Architecture Diagram	Chapter 3
3.2	Data Flow Diagram – Level 0 (Context Diagram)	Chapter 3
3.3	Data Flow Diagram – Level 1	Chapter 3
3.4	Data Flow Diagram – Level 2 (Document Management Module)	Chapter 3
3.5	Control Flow Diagram	Chapter 3
3.6	State Transition Diagram	Chapter 3
3.7	Use Case Diagram	Chapter 3
3.8	Entity Relationship (ER) Diagram	Chapter 3
5.1	UI Screenshot – Dashboard Screen	Chapter 5
5.2	UI Screenshot – Health Documents Screen	Chapter 5
5.3	UI Screenshot – Appointments Screen	Chapter 5
5.4	UI Screenshot – Medications Screen	Chapter 5

List of Tables

Table No.	Table Title	Chapter
1.1	Definitions and Abbreviations	Chapter 1
2.1	Functional Requirements Summary	Chapter 2
2.2	Non-Functional Requirements	Chapter 2
3.1	Database Collections Description	Chapter 3
4.1	Technology Stack Summary	Chapter 4
4.2	Test Cases – Authentication Module	Chapter 4
4.3	Test Cases – Document Management Module	Chapter 4
4.4	Test Cases – Appointment Module	Chapter 4
5.1	User Collection Fields	Chapter 5
5.2	Document Collection Fields	Chapter 5
5.3	Appointment Collection Fields	Chapter 5

Abbreviations

Abbreviation	Full Form
SRS	Software Requirements Specification
PHR	Personal Health Record
EHR	Electronic Health Record
DFD	Data Flow Diagram
ER	Entity Relationship
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
API	Application Programming Interface
REST	Representational State Transfer
HTTPS	HyperText Transfer Protocol Secure
CRUD	Create, Read, Update, Delete
UI	User Interface
UX	User Experience

JWT	JSON Web Token
DB	Database
IDE	Integrated Development Environment
WHO	World Health Organization
ABDM	Ayushman Bharat Digital Mission

Notations & Symbols

Symbol / Notation	Description
REQ-X.Y	Requirement identifier: X = module number, Y = requirement number
UC-N	Use Case identifier, N = sequential use case number
P1 – P7	Process identifiers for the seven pipeline modules
D1 – D5	Data store identifiers in DFD Level 1
[]	Denotes a set or collection of items
{ }	Denotes a JSON object or structured data record
→	Data flow direction in DFD notation
TBD-N	To-Be-Determined item, N = sequential number
NFR-XX	Non-Functional Requirement identifier
FR-X.Y	Functional Requirement identifier (same as REQ-X.Y)
DXA	Twentieth of a point; unit of measurement in OOXML document format
HTTP 4xx	Client-side HTTP error response code family
HTTP 5xx	Server-side HTTP error response code family

CHAPTER 1 INTRODUCTION

1.1 Introduction

The healthcare sector is one of the most essential domains, directly influencing human life and well-being. However, managing patient information efficiently remains a significant challenge for many healthcare institutions. Traditionally, hospitals and clinics have relied on paper-based systems for storing medical records, scheduling appointments, and maintaining patient data. While this approach has been used for decades, it often leads to issues such as difficulty in accessing records, risk of misplacement or damage of documents, and inefficiencies in handling appointments and patient flow.

With the advancement of technology, the integration of digital solutions into healthcare has become increasingly important. Modern web-based systems provide a more efficient and reliable way to manage healthcare data. In this context, a **Centralized Web-Based Health Document and Appointment Management System** is proposed to streamline the process of managing patient information and healthcare activities.

The primary objective of this system is to provide a unified digital platform where users can store and manage their medical documents securely. Patients can upload prescriptions, diagnostic reports, and other health-related records, eliminating the need to carry physical copies during hospital visits. This not only reduces inconvenience but also minimizes the risk of losing important documents. The system also supports multiple user profiles, allowing families to manage the health records of different members within a single account.

Another important feature of the system is the appointment management module. In many healthcare facilities, patients often face long waiting times and difficulties in scheduling appointments. The proposed system simplifies this process by allowing users to schedule appointments and set reminders for upcoming visits, vaccinations, or other health-related events. This helps in improving time management and reducing delays.

Additionally, the system includes a medication tracking feature that enables users to record their medications, dosage schedules, and refill dates. This functionality helps users maintain consistency in their treatment and reduces the chances of missing doses.

Data security and privacy are critical aspects of any healthcare system. Since medical records contain sensitive information, the system incorporates secure authentication methods,

encrypted data storage, and role-based access control to ensure that only authorized users can access the data. This ensures both safety and confidentiality of patient information.

Furthermore, the centralized nature of the system allows easy access to medical records from different locations when needed. This can be especially useful in emergency situations, where immediate access to a patient's medical history can assist in accurate and timely treatment.

In conclusion, the proposed system aims to improve healthcare management by providing a secure, efficient, and user-friendly platform for managing health documents, appointments, and medications. By replacing traditional paper-based methods with a digital solution, it enhances organization, accessibility, and overall healthcare experience for users.

Table 1.1: Definitions and Abbreviations

Term	Description
SRS	Software Requirements Specification
User	Any person using the system
Parent	User who can manage own and child profiles
Child	Profile managed by a parent
Document	Digital file such as PDF or image stored in the system
Reminder	Notification for an upcoming appointment or event
Authentication	Process of verifying user identity
PHR	Personal Health Record
EHR	Electronic Health Record
Medication	Prescription drug being tracked by the user

1.2 Literature Review

Introduction to Literature Review

Hospitals, laboratories, pharmacies, and diagnostic centres have progressively adopted Electronic Health Records (EHRs) with the ongoing digitalization of healthcare. While EHRs lead to improvements in clinical documentation and operational efficiency, patient medical documents remain scattered across various organizations and providers, limiting clinical decision making, decreasing continuity of care, and increasing the administrative burden on patients and their families.

Individuals can access, store, and share their medical history and documents with the help of Personal Health Records (PHRs), while personally managing the platform according to their needs. However, PHR systems are mainly focused on report viewing and only offer basic appointment booking functionality, leading to limited scope and functionality for comprehensive family health management.

Emerging research highlights the need for patient and family-oriented platforms that support comprehensive medical documentation, tracking of appointments with reminders, medication management, and management of multiple profiles such as parents, their children, and wards, all under a single user account.

Personal Health Records (PHR) Systems

PHRs are safe, patient-controlled systems that let people manage and share their medical records. According to research, contemporary PHRs have evolved from static data stores to interactive platforms that facilitate patient-provider communication, appointment scheduling, and limited self-monitoring functions. However, because of their limited capability, lack of connectivity with numerous data sources, and limited customization, the majority of deployed systems still have little effect on long-term healthcare management. Tang et al. (2006) emphasized that PHR adoption requires overcoming barriers related to usability, trust, and integration with existing healthcare systems. Roehrs et al. (2017) conducted a systematic literature review of 1534 articles and found that the most common PHR functionalities are access to medical records, medication management, and communication with providers.

Health Document Management Systems

Patients are frequently forced to manually retain prescriptions, laboratory reports, diagnostic imaging, and vaccination records. Literature on health document management highlights how

medical data fragmentation among several healthcare providers leads to incomplete medical histories, repetitive diagnostic testing, and ineffective treatment planning. Centralized digital repositories that enable safe storage, classification, retrieval, and restricted exchange of medical records are strongly supported by research. Structured data models and interoperability standards significantly improve integration among healthcare systems. The work of Blumenthal and Tavenner (2010) on Meaningful Use regulation for EHRs emphasized the importance of electronic access, storage, and exchange of health information for improving the quality and efficiency of healthcare.

Family-Centred and Multi-Profile Health Management

According to recent studies, healthcare management involves more than just individuals; it also involves families and caregivers, particularly in the areas of pediatric, elder, and chronic illness care. Managing dependent profiles, linked records, and role-based access control are examples of family-based workflows that are not supported by the majority of PHR and EHR systems currently in use, since they are designed with the individual in mind. Gabbay et al. (2022) proposed family-centered digital health systems that preserve auditability, privacy, and access control while enabling a principal user to oversee several profiles. This is particularly relevant for pediatric health management where parents must coordinate healthcare for multiple children.

Appointment Management and Reminder Systems

Missed or delayed appointments are one of the main causes of poor treatment adherence and health outcomes. Reminder systems have been shown to dramatically increase appointment attendance and treatment continuity. A large number of PHR platforms currently in use either do not have reminder features or only offer simple notifications without integrating profilebased health records and document management. Research indicates that integrated appointment management with automatic reminders is one of the most important features for increasing user engagement and healthcare adherence. Studies by healthcare technology experts confirm that web-based and mobile reminder systems can reduce missed appointment rates by up to 36%.

Medication Tracking and Management

Medication non-adherence is a major challenge in healthcare, leading to worsening of chronic conditions, hospitalization, and increased healthcare costs. Digital medication tracking systems that allow patients to record their medications, dosage schedules, and refill reminders have been shown to improve adherence rates. Studies indicate that patients with access to medication management tools are significantly more likely to follow their prescribed treatment regimes.

However, most existing PHR platforms treat medication tracking as a secondary feature rather than integrating it as a core component of the health management platform.

Key Challenges Identified in Existing Systems

- **Data Fragmentation:** Patient records scattered across multiple platforms and providers lead to incomplete medical histories.
- **Limited Patient Control:** Few alternatives are available to patients for controlling document organization and access privileges.
- **Poor Interoperability:** Inconsistent data standards make it difficult for medical records to be seamlessly integrated and exchanged.
- **Issues with Usability and Adoption:** Complex user interfaces make it difficult for users to engage effectively.
- **Privacy and Security Concerns:** Sensitive health data requires robust authorization, encryption, and authentication processes.
- **Lack of Family-Centric Design:** Existing platforms do not adequately support management of multiple dependent profiles.
- **Inadequate Medication Management:** Most systems lack integrated medication tracking and refill reminder features.
- **Information Overload:** Organizing and retrieving large volumes of unstructured documents proves difficult.

Limitations in Existing Research and Implementation

Most existing research is literature-based and may not be representative of real-world deployment problems. Many systems are designed around healthcare professionals rather than patients and families. The regional focus of research limits its applicability to different healthcare infrastructures and regulatory frameworks. Additionally, on existing platforms, data access usually takes precedence over proactive health management features such as role-based profile management, intelligent organization, medication tracking, and appointment reminders.

Research Gap and Motivation for Proposed System

The gap between existing digital health systems and the practical needs of patients and their families is evident from the literature. The following capabilities are not commonly combined in a single system: centralized classification and storage of documents; automated appointment scheduling and reminders; medication tracking with refill notifications; management of multiple family members' profiles; and robust user-controlled access, privacy, and security measures. This gap is addressed by the proposed consolidated web-based system that unifies these various functions into a single platform, enhancing accessibility, organization, and compliance with healthcare standards.

Conclusion of Literature Review

Despite significant developments, PHRs and health document management systems remain fragmented, individual-centric, and functionally limited. A single, patient-controlled, family-oriented web-based platform that integrates medical document management, appointment scheduling, medication tracking, and reminders is clearly needed. Such a system can improve continuity of treatment, increase patient participation, reduce administrative burden, and assist individuals and families in better managing their healthcare.

1.3 Objectives

The main objectives of the proposed Centralized Web-Based Health Document and Appointment Management System are as follows:

1. To design and develop a secure, centralized web-based platform for storing and managing personal health documents for individuals and families.
2. To provide a unified digital repository where users can upload, view, download, delete, and categorize health documents such as prescriptions, diagnostic reports, laboratory results, vaccination records, and treatment histories.
3. To support multi-profile management, enabling parents to create and manage child profiles under a single account with appropriate role-based access control.
4. To implement a comprehensive appointment management module allowing users to schedule, edit, and delete appointments along with automated reminder notifications.
5. To provide a medication tracking module enabling users to record current medications, dosage schedules, and set refill reminders.

6. To ensure data security and privacy through secure authentication, password encryption, HTTPS communication, and role-based access control mechanisms.
7. To provide a responsive and user-friendly interface accessible across desktop, tablet, and mobile devices on modern web browsers.
8. To reduce reliance on paper-based medical record keeping, thereby minimizing the risk of document loss, damage, or unauthorized access.
9. To improve healthcare management efficiency by enabling authorized healthcare providers to access complete patient medical histories during consultations.

1.4 Significance

The Centralized Web-Based Health Document and Appointment Management System holds significant importance for individuals, families, and the broader healthcare ecosystem.

For individual patients and families, the system provides a secure, convenient platform to maintain organized digital health records. This eliminates the need to carry physical documents to medical appointments, reducing the risk of document loss and enabling healthcare providers to make more informed decisions. For families managing the health of children or elderly dependents, the multi-profile feature simplifies the coordination of medical information across multiple family members.

From a public health perspective, centralized health record management supports continuity of care across different healthcare providers. When patients visit multiple hospitals or clinics, their complete medical history remains accessible, reducing the likelihood of repeated diagnostic tests and the associated costs.

The appointment reminder feature directly addresses the problem of missed medical appointments, which is a significant contributor to poor health outcomes and wasted medical resources. Studies have consistently shown that reminder-based systems improve appointment adherence, particularly for preventive care and chronic disease management.

The medication tracking feature addresses a critical healthcare challenge. Medication nonadherence is responsible for approximately 125,000 deaths annually in the United States alone.

By providing a structured, easily accessible record of medications along with refill reminders, the system directly supports better health outcomes.

The system also contributes to digital literacy and patient empowerment by encouraging individuals to actively participate in managing their own health information. By making health records easily accessible and understandable, the system supports informed decision-making and improved communication between patients and healthcare providers.

1.5 Research Design / Methodology

The development of the Centralized Web-Based Health Document and Appointment Management System follows a structured Software Development Life Cycle (SDLC) methodology.

Phase 1: Requirement Analysis

In this phase, the needs and expectations of users were carefully studied and documented. Data was gathered through research on existing healthcare systems, analysis of similar digital healthcare platforms, and review of related literature. Both functional requirements (what the system should do) and non-functional requirements (how the system should perform) were identified.

Phase 2: System Design

Following requirement analysis, the system design phase focused on defining the overall architecture and structure of the application. This included creating Data Flow Diagrams (DFD) at levels 0, 1, and 2, Entity Relationship (ER) Diagrams, Use Case Diagrams, State Transition Diagrams, and Control Flow Diagrams. The database schema was designed to efficiently store user profiles, documents, appointment records, medication data, and notification data.

Phase 3: Technology Selection

Appropriate technologies were selected to ensure reliability, scalability, and security. The frontend was developed using HTML, CSS, JavaScript, and React.js. The backend was implemented using Node.js and Express.js. MongoDB was selected as the database management system for its flexibility in handling document-based data structures.

Phase 4: Implementation

The implementation phase involved converting the system design into a working application. Development was carried out in stages: user authentication, document management module, appointment management module, medication tracking module, and profile management features. Coding standards and best practices were followed throughout.

Phase 5: Testing

Multiple testing methodologies were applied to verify correct system functionality. Unit testing validated individual modules, integration testing confirmed interaction between modules, system testing evaluated overall system performance, and security testing verified data protection mechanisms.

1.6 Source of Data

The information for this project was gathered from multiple reliable sources to ensure comprehensive coverage of the domain:

- Academic journals and research papers on Personal Health Records (PHRs), Electronic Health Records (EHRs), and family-centred healthcare management.
- Published books on software engineering, web development, and healthcare information systems.
- Official documentation for technologies used including React.js, Node.js, Express.js, and MongoDB.
- World Health Organization (WHO) publications on digital health initiatives and healthcare digitalization.
- Government of India digital health initiatives reports and guidelines from the Ministry of Health and Family Welfare.

- IEEE publications, medical informatics journals, and technology documentation websites.
- Analysis of existing healthcare management systems and digital health platforms to identify gaps and opportunities for improvement.

1.7 Chapter Scheme

This report is organized into seven chapters, as described below:

- Chapter 1 - Introduction: Provides an overview of the project, literature review, problem statement, objectives, significance, methodology, and chapter outline.
- Chapter 2 - Requirements Specification: Describes the functional and non-functional requirements of the system, user characteristics, dependencies, hardware and software requirements, and constraints and assumptions.
- Chapter 3 - System Design: Presents the system architecture and all design diagrams including Data Flow Diagrams, Control Flow Diagram, State Transition Diagram, Use Case Diagram, ER Diagram, and Database Design.
- Chapter 4 - Implementation, Testing, and Maintenance: Details the technologies and tools used, testing techniques and test plans, installation instructions, and end-user instructions.
- Chapter 5 - Results and Discussions: Describes the modules of the system, user interface screenshots, and database tables.
- Chapter 6 - Summary and Conclusions: Summarizes the work done, presents conclusions, and discusses the project's contributions.
- Chapter 7 - Future Scope: Identifies potential enhancements and extensions for future versions.
- Bibliography: Lists all references cited in the report.

CHAPTER 2 REQUIREMENTS SPECIFICATION

2.1 User Characteristics

The system serves two primary types of users, each with distinct roles, responsibilities, and access levels:

Parent User

The parent user is the primary account holder who registers with the system using an email address and password. Once authenticated, the parent user has the highest level of access within their account and can perform the following actions:

- Register and manage their own personal profile including name, date of birth, contact details, and optional profile photo.
- Create and manage child profiles linked to their parent account.
- Upload, view, download, and delete health documents for both their own and their children's profiles.
- Categorize documents into predefined categories such as Medical Records, School Documents, Vaccination Records, Prescriptions, and Diagnostic Reports.
- Schedule, edit, and delete appointments and receive reminder notifications for all linked profiles.
- Add, edit, and track medications including dosage, frequency, and refill dates.
- Search and retrieve documents by name or category and apply document tags.

Child User

The child user is a dependent profile created and managed by a parent user. Child users have limited, profile-restricted access:

- Access and view documents stored under their own profile.
- View reminders and appointments scheduled for their profile.
- View medications recorded for their profile.
- Cannot access or modify documents belonging to other profiles.

Operating Environment

- Web browsers: Google Chrome, Microsoft Edge, Mozilla Firefox, and Apple Safari.
- Operating systems: Windows, Android, iOS, and macOS.
- Devices: Desktop computers, laptops, tablets, and mobile phones.
- An active internet connection is required to access all system features.

2.2 Functional Requirements

FR-1: User Registration and Login

- The system shall allow users to create an account using a valid email address and password.
- The system shall validate email format and enforce minimum password complexity requirements.
- The system shall support secure login and logout functionality.
- The system shall support multiple profile types: parent and child.
- Parents shall be able to create and manage child profiles under their account.
- The system shall use JWT (JSON Web Tokens) for session management.

FR-2: Profile Management

- Users shall be able to create, view, edit, and delete their profiles.
- Parents shall be able to add multiple children under their account.
- Each profile shall store name, date of birth, contact details, and optional profile photo.
- The system shall maintain separate document, appointment, and medication records for each profile.

FR-3: Document Management

- Users shall be able to upload documents in PDF, JPEG, PNG, and other common formats.
- Users shall be able to view, download, and delete uploaded documents.
- Documents shall be categorized under predefined categories including Aadhaar, Birth Certificate, Medical Records, School Documents, Vaccination Records, Prescriptions, and Diagnostic Reports.
- Parents shall be able to manage documents for their children's profiles.

FR-4: Document Organization

- The system shall allow documents to be organized into folders or categories.
- The system shall provide a search function to find documents by name, category, or date.
- The system shall allow document tagging for easy identification and retrieval.

FR-5: Appointment Reminder System

- Users shall be able to add appointments including doctor visits, school meetings, and vaccination schedules.
- Users shall be able to set a date and time for each appointment and configure reminder notifications.
- The system shall generate email reminder notifications before the appointment date and time.
- Users shall be able to view upcoming appointments on a calendar and in list view.
- Users shall be able to reschedule or cancel existing appointments.

FR-6: Medication Tracking

- Users shall be able to add medications with name, dosage, frequency, and start date.
- Users shall be able to record refill dates and request refills.
- The system shall display active medications with dosage and scheduling information.
- Users shall be able to edit medication details and remove medications no longer in use.

FR-7: Access Control

- Parent users shall have full access to their own profile and all linked child profiles.
- Child users shall only have access to their own profile and documents.
- Unauthorized users shall not be able to access other users' documents or profiles.

FR-8: Notifications

- The system shall send reminder notifications for upcoming appointments.
- The system shall notify users when documents are successfully uploaded or updated.
- Notifications shall be displayed on the user dashboard and optionally sent via email.

Table 2.1: Functional Requirements Summary

Req. ID	Requirement	Priority	User
FR-1	User Registration and Login	High	All Users

FR-2	Profile Management	High	Parent, Child
FR-3	Document Management	High	Parent, Child
FR-4	Document Organization	Medium	Parent, Child
FR-5	Appointment Reminder System	High	Parent, Child
FR-6	Medication Tracking	High	Parent, Child
FR-7	Access Control	High	All Users
FR-8	Notifications	Medium	All Users

2.3 Dependencies

- Frontend: React.js, React Router, Axios.
- Backend: Node.js, Express.js, Mongoose, Multer, bcrypt, JWT, NodeMailer.
- Database: MongoDB / MongoDB Atlas.
- Cloud Storage: Cloudinary or AWS S3 for uploaded document files.
- Internet connectivity for both users and server operations.

2.4 Performance Requirements

Table 2.2: Non-Functional Requirements

Category	Requirement
----------	-------------

Performance	Pages shall load within 2-3 seconds. Document operations shall complete within acceptable time based on file size.
Reliability	System shall be available 24/7. Data shall be protected through backups. System shall recover from failures promptly.
Security	All communications shall use HTTPS. Passwords shall be bcrypt-hashed. Access shall be controlled via JWT and role-based authorization.
Usability	Interface shall be simple and intuitive. Users shall navigate without external guidance. Navigation shall be clear and consistent.
Scalability	System shall support an increasing number of users and growing document volumes efficiently.
Compatibility	System shall function on Chrome, Edge, Firefox, and Safari. Interface shall be fully responsive on all screen sizes.
Maintainability	Codebase shall follow standard practices. Bug fixes and feature additions shall be implementable without complete redesign.

2.5 Hardware & Software Requirements

Server Side Hardware

- Processor: Multi-core, minimum 2 GHz.
- RAM: 4 GB minimum (8 GB for production).
- Storage: Adequate for OS, application, and database.
- Network: Stable internet with sufficient bandwidth.

Client Side Hardware

- Any modern desktop, laptop, tablet, or smartphone with internet access.
- Minimum 360 x 640 px (mobile) or 1024 x 768 px (desktop) screen resolution.

Server Side Software

- OS: Linux (Ubuntu recommended), Windows Server, or macOS.
- Node.js v16+, Express.js v4+, MongoDB v5+, npm.

Client Side Software

- Web Browser: Chrome (v90+), Firefox (v88+), Edge (v90+), Safari (v14+).
- JavaScript must be enabled. PDF viewer required for document preview.

2.6 Constraints & Assumptions

Constraints

- Active internet connection required. Offline functionality not supported in version 1.
- Users must register before accessing system features.
- System must ensure secure authentication at all times.
- Document file sizes limited to 10 MB per file.
- Supported formats: PDF, JPEG, PNG, DOCX.

Assumptions

- Users have basic knowledge of internet and computer or smartphone usage.
- Users will upload valid, legitimate health-related documents.
- Server, database, and cloud storage will maintain acceptable uptime.
- Email reminders are assumed to be successfully delivered subject to the user's email service availability.

CHAPTER 3 SYSTEM DESIGN

3.1 System Architecture

The system follows a three-tier client-server architecture separating the Presentation Layer (Frontend), Application Logic Layer (Backend), and Data Layer (Database). Each layer communicates through well-defined REST API interfaces, ensuring modularity, scalability, and ease of maintenance.

Presentation Layer (Frontend)

Developed using React.js, this layer renders the user interface and handles all user interactions. React Router handles client-side navigation between sections without full page reloads. Axios

handles HTTP communication with the backend REST API. The interface is responsive and works across all modern devices and browsers.

Application Logic Layer (Backend)

Developed using Node.js with Express.js, this layer processes all business logic, validates requests, manages authentication using JWT tokens, handles file uploads via Multer, secures passwords with bcrypt, and sends email notifications via NodeMailer.

Data Layer (Database)

MongoDB, a document-oriented NoSQL database, stores all application data including user profiles, document metadata, appointment records, medication data, and notification history. The database is accessed exclusively through the backend API, preventing any direct database access from the client.

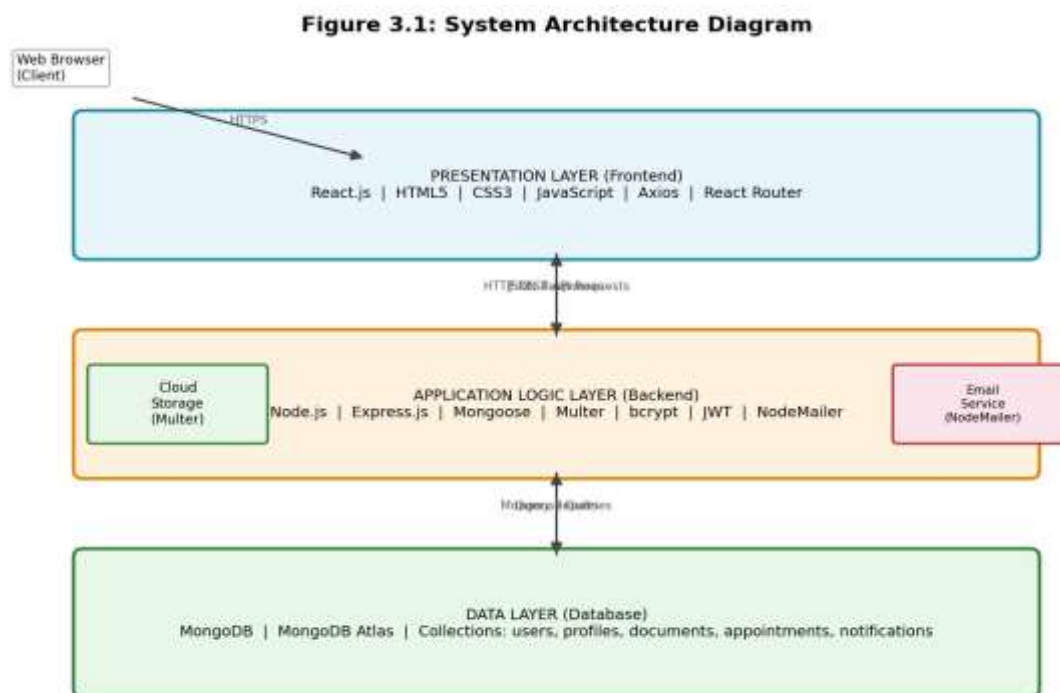


Figure 3.1: System Architecture Diagram

3.2 Data Flow Diagrams

Level 0: Context Diagram

The Level 0 DFD represents the entire system as a single process and defines the system boundary. External entities include the User (Parent/Child) and the Email Service. Data flows into the system include login credentials, registration data, document files, profile management data, and appointment details. Data flows out include document access, notifications, authentication responses, and dashboard data.

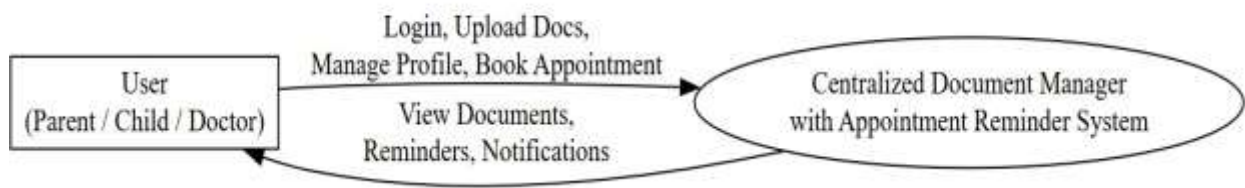


Figure 3.2: Data Flow Diagram – Level 0 (Context Diagram)

Level 1: DFD Level 1

The Level 1 DFD breaks the system into its main functional modules. Five primary processes are identified: (1.0) User Authentication, (2.0) Profile Management, (3.0) Document Management, (4.0) Appointment Management System, and (5.0) Notification System. These processes interact with three data stores: User DB, Document DB, and Appointment DB.

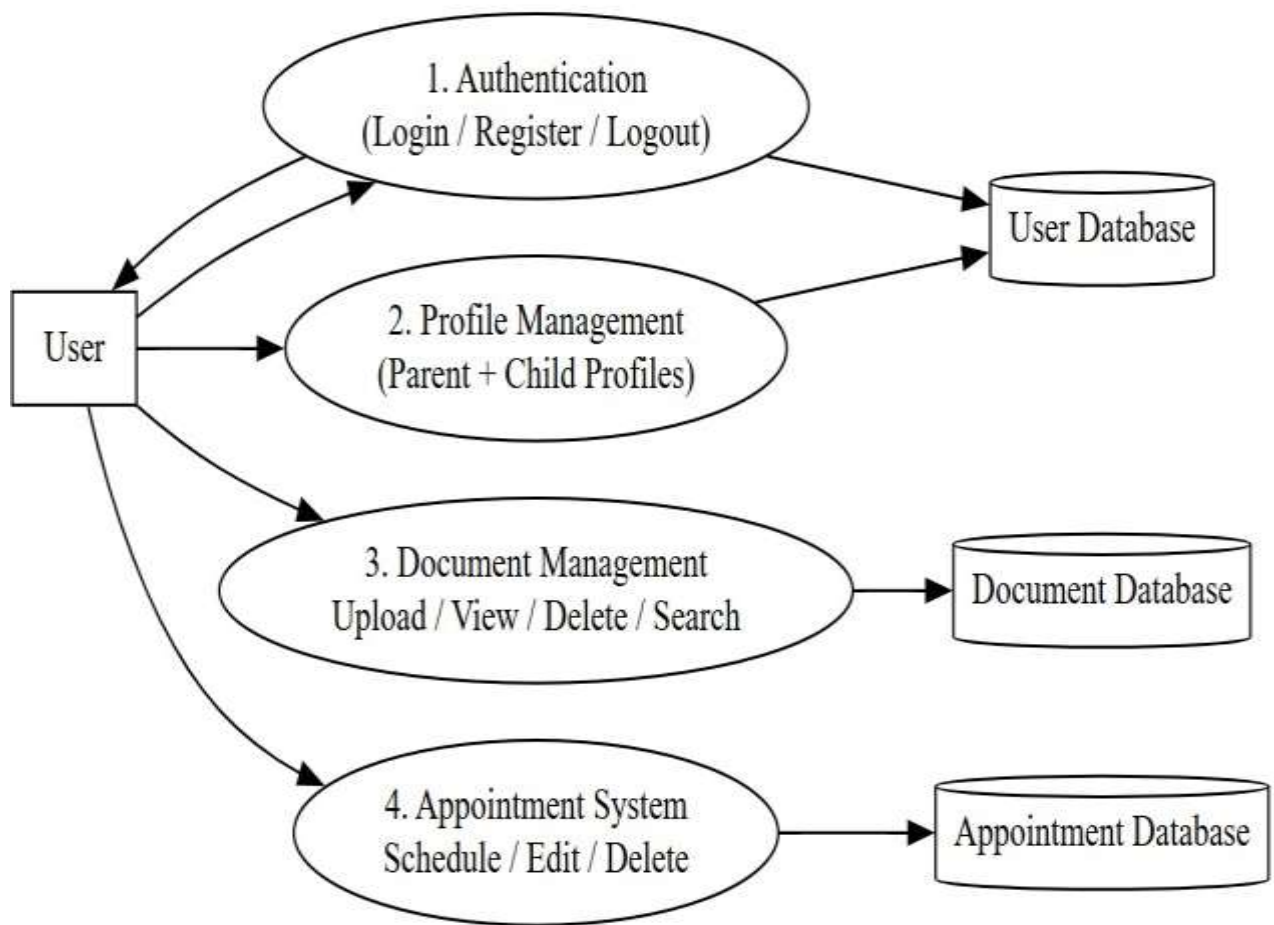


Figure 3.3: Data Flow Diagram – Level 1

Level 2: DFD Level 2 – Document Management Module

The Level 2 DFD provides a detailed decomposition of the Document Management module. Sub-processes include 3.1 Validate & Upload, 3.2 View Documents, 3.3 Download Document, 3.4 Delete Document, and 3.5 Search & Categorize. These processes interact with the Document DB and Cloud Storage external entity.



Figure 3.4: Data Flow Diagram – Level 2 (Document Management Module)

3.3 Control Flow Diagram

The Control Flow Diagram describes the logical sequence of operations and decision points as users interact with the system. The flow begins with the user opening the application, proceeding through registration or login, credential validation, dashboard access, and module navigation, and terminating with logout.

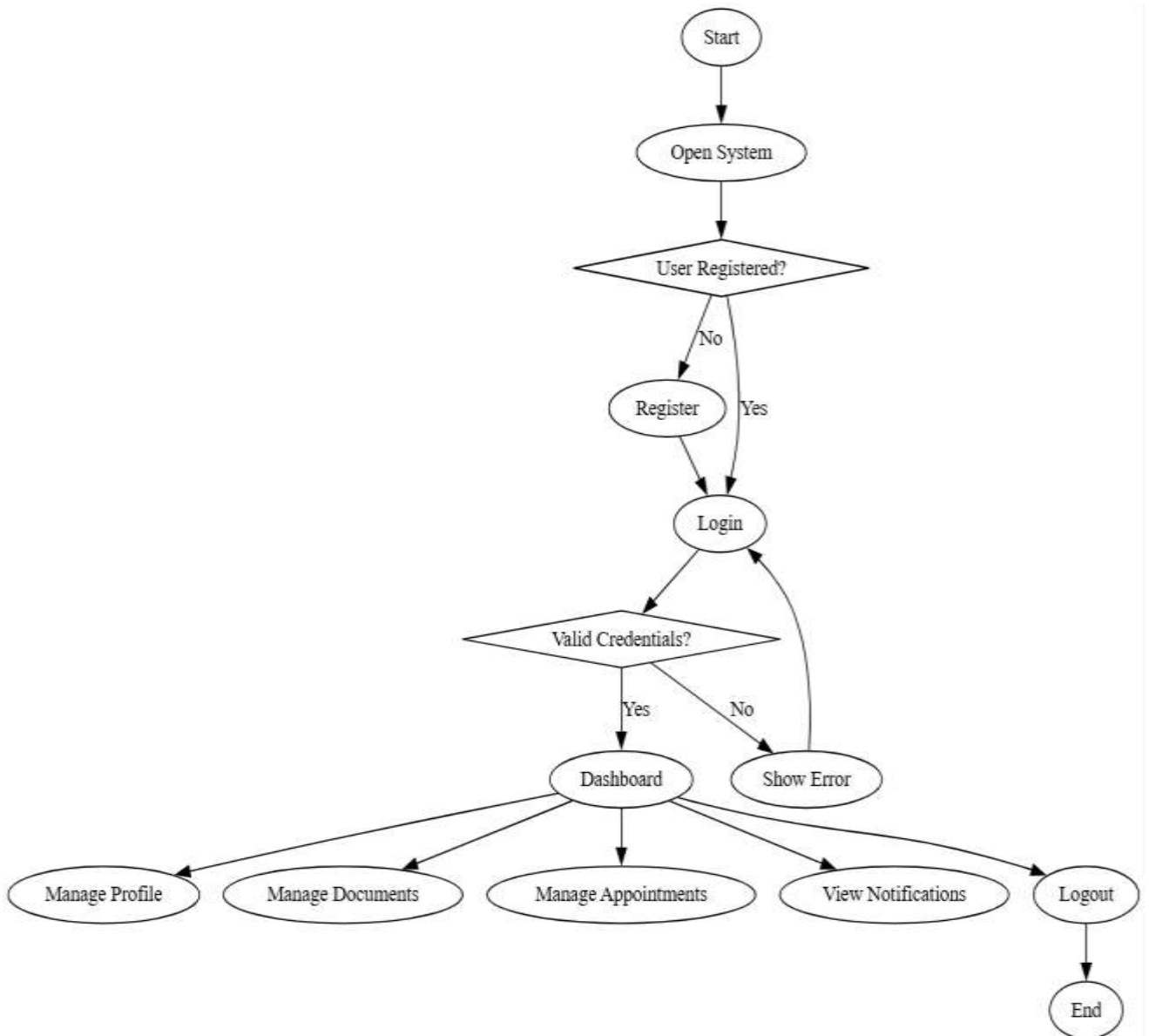


Figure 3.5: Control Flow Diagram

3.4 State Transition Diagram

The State Transition Diagram shows the different states of the system and transitions based on user actions or system events. The system begins in the Unauthenticated State and transitions to the Authenticated Dashboard State on successful login. From the dashboard, users navigate to Profile, Document, Appointment, or Notification states. Logout or session timeout returns the system to the Unauthenticated State.

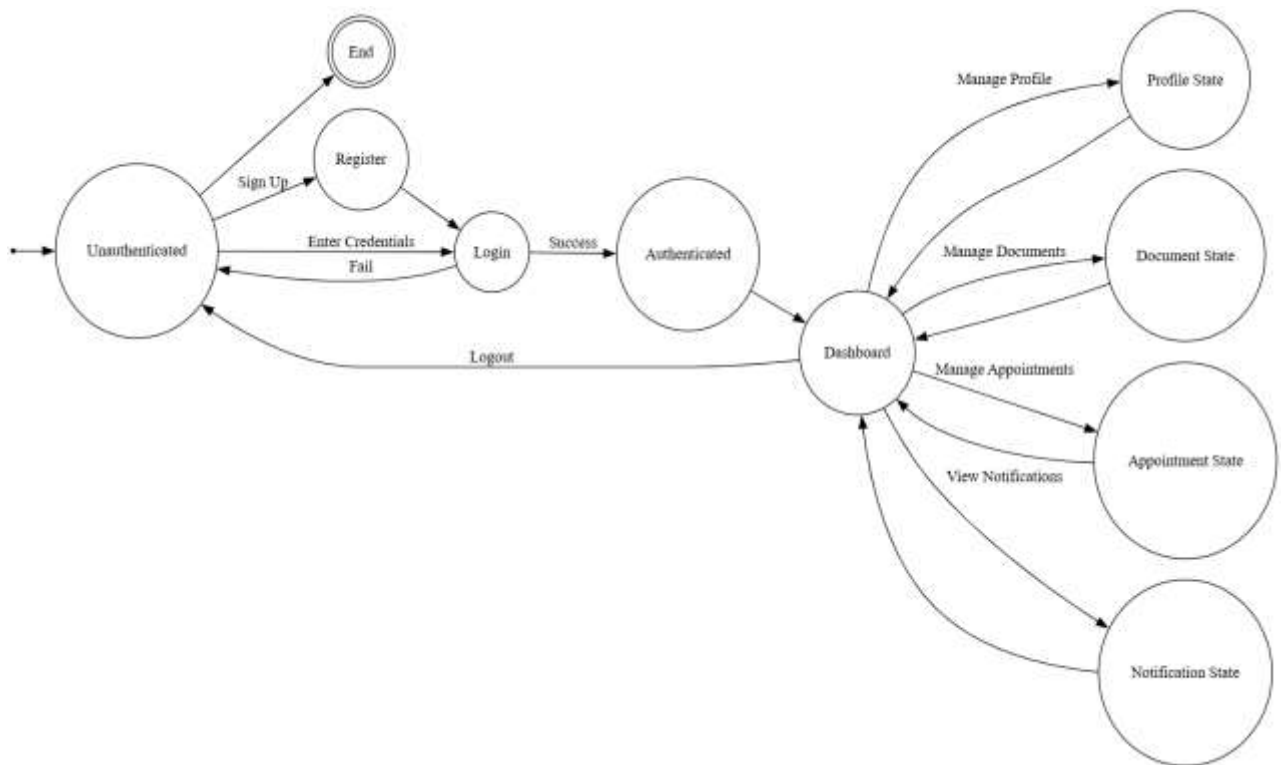


Figure 3.6: State Transition Diagram

3.5 Use Case Diagram

The Use Case Diagram illustrates the functional interactions between actors and the system. Actors include Parent User, Child User, and Email Service. The system boundary encompasses all application use cases. Parent users have full access to all use cases while child users have restricted access to their own profile's functionalities. The Email Service actor is linked to the Receive Notification and Send Email Reminder use cases.

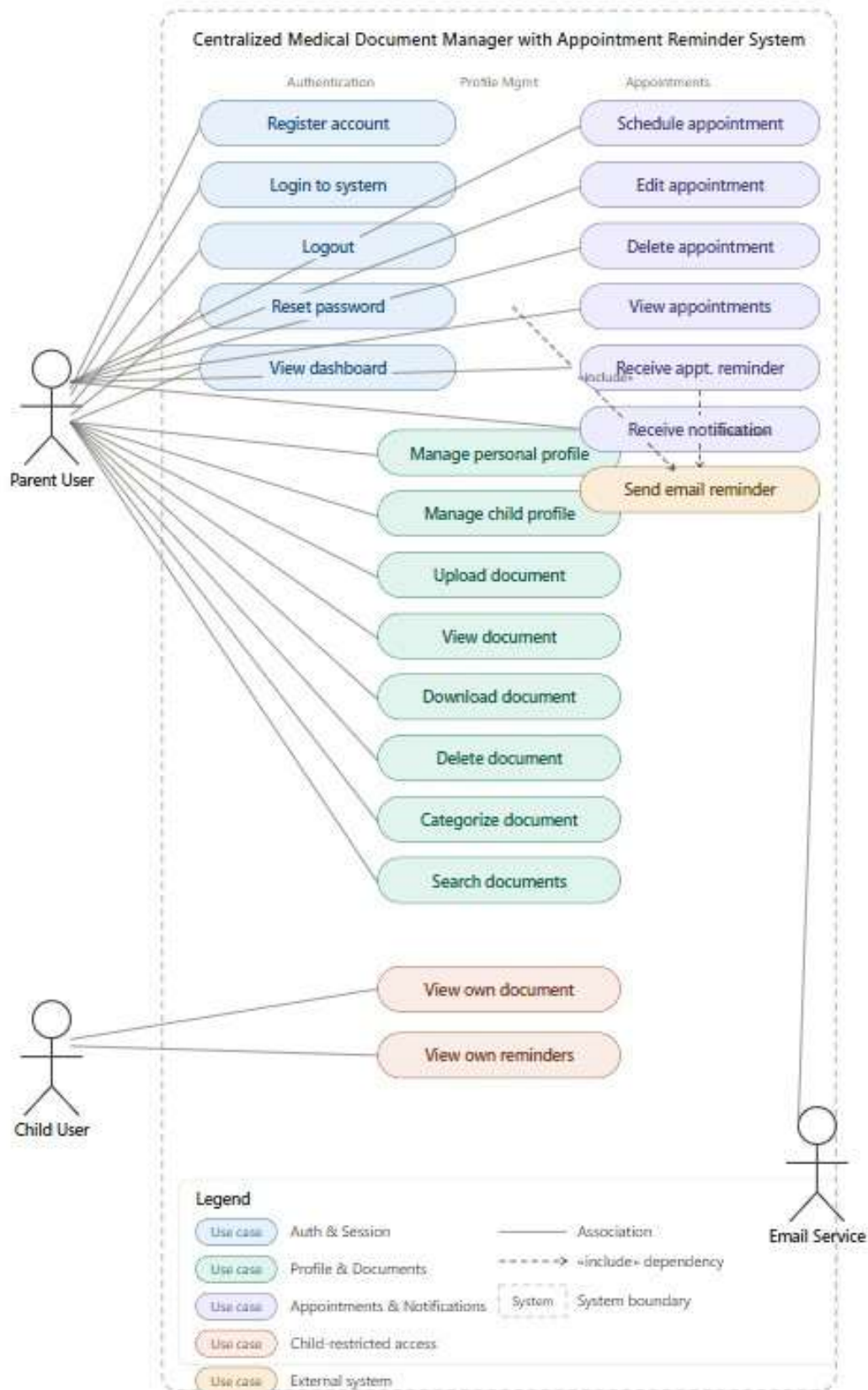


Figure 3.7: Use Case Diagram

3.6 Entity Relationship (ER) Diagram

The ER Diagram represents the database structure with main entities including User, Profile, Document, Category, Appointment, Notification, and Audit Log. Key relationships are: User to Profile (1:N), Profile to Document (1:N), Document to Category (N:1), Profile to Appointment (1:N), Appointment to Notification (1:N), and User to Audit Log (1:N).

Figure 3.8: Entity Relationship (ER) Diagram

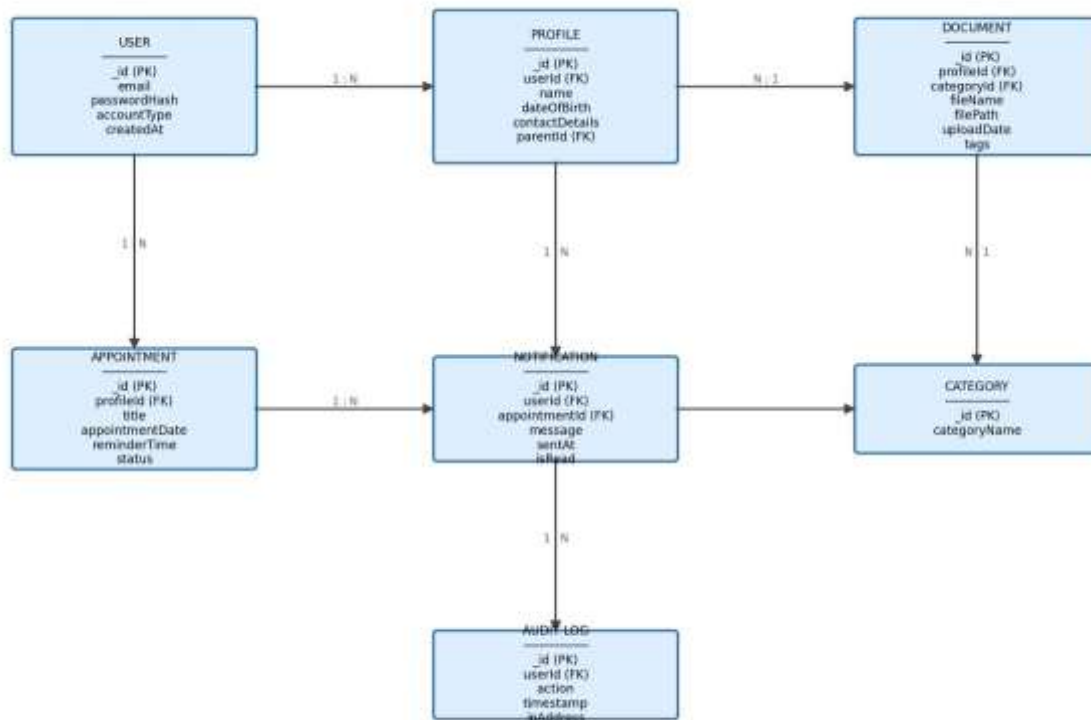


Figure 3.8: Entity Relationship (ER) Diagram

3.7 Database Design

Logical Database Design

The system uses MongoDB with data organized into collections. The logical design includes the following collections:

Table 3.1: Database Collections Description

Collection	Description	Key Fields
users	Stores user account data including email, hashed password, and account type.	_id, email, passwordHash, accountType
profiles	Stores parent and child profile data linked to user accounts.	_id, userId, name, dob, parentId
documents	Stores document metadata including file references, categories, and tags.	_id, profileId, categoryId, fileName, filePath, tags
categories	Stores predefined document category names.	_id, categoryName
appointments	Stores appointment records with date, time, type, and reminder settings.	_id, profileId, title, date, reminderTime, status
medications	Stores medication records with dosage, frequency, and refill dates.	_id, profileId, name, dosage, frequency, refillDate
notifications	Stores notification history for sent reminders and system alerts.	_id, userId, appointmentId, message, sentAt, isRead
auditlogs	Stores audit records of user actions for security monitoring.	_id, userId, action, timestamp, ipAddress

Physical Database Design

The database is implemented on MongoDB (Atlas for production). Indexes are applied on email in users collection, profileId and categoryId in documents, profileId and appointmentDate in

appointments, and `userId + isRead` in notifications. Data validation is enforced using Mongoose schema definitions on the backend.

CHAPTER 4 IMPLEMENTATION, TESTING, AND MAINTENANCE

4.1 Technologies Used

The system was developed using a modern, full-stack JavaScript technology stack selected for scalability, security, and developer productivity.

Frontend Technologies

- HTML5: Defines semantic structure of all web pages.
- CSS3: Applied for styling, layout, animations, and visual consistency.
- JavaScript (ES6+): Client-side interactivity, form validation, dynamic content.
- React.js: Component-based library for building the dynamic, responsive UI. Virtual DOM enables efficient rendering.
- React Router: Client-side navigation without full page reloads.
- Axios: HTTP client for making API calls from frontend to backend.
- Tailwind CSS / Bootstrap: Responsive layout and pre-built UI components.

Backend Technologies

- Node.js: Server-side JavaScript runtime providing non-blocking, event-driven architecture.
- Express.js: Web framework for RESTful API endpoint creation and middleware management.
- Mongoose: ODM library providing schema-based data modelling and validation for MongoDB.
- Multer: Middleware for handling multipart/form-data for document file uploads.
- bcrypt: Library for secure password hashing and comparison.
- JSON Web Tokens (JWT): Stateless authentication through secure token generation and verification.
- NodeMailer: Module for sending appointment reminder emails to users.

Database Technologies

- MongoDB: NoSQL document-oriented database for all application data storage.
- MongoDB Atlas: Cloud-hosted MongoDB service for production deployment with automated backups.

Table 4.1: Technology Stack Summary

Layer	Technology	Purpose
Frontend	React.js, HTML5, CSS3, JavaScript	User Interface Development
Routing	React Router	Client-Side Navigation
HTTP Client	Axios	API Communication
Backend	Node.js, Express.js	Server & API Development
Authentication	JWT, bcrypt	Security & Session Mgmt.
File Upload	Multer	Document Upload Handling
Email	NodeMailer	Reminder Notifications
Database	MongoDB, Mongoose	Data Persistence
Cloud DB	MongoDB Atlas	Production DB Hosting

Development Tools and IDE

- Visual Studio Code (VS Code): Primary IDE for all code writing and debugging.
- Postman: Testing and debugging RESTful API endpoints.
- Git + GitHub: Version control and remote repository hosting.
- npm: Dependency management and development scripts.

4.2 Testing Techniques and Test Plans

Testing ensured the system meets its specified requirements and functions correctly under all expected conditions.

Unit Testing

Individual functions and modules were tested in isolation using the Jest testing framework. Each API endpoint, utility function, and React component was tested with specific input values to confirm expected output.

Integration Testing

Integration testing verified that different modules work correctly when combined. This included frontend-backend API interactions, backend-database operations, and notification system with email service. Postman was used extensively.

System Testing

System testing evaluated the complete integrated system against all functional and nonfunctional requirements. Test cases covered all user scenarios from registration through document management, appointment scheduling, and medication tracking.

Security Testing

Security testing verified authentication mechanisms, access control, JWT token expiry, and bcrypt password hashing. Tests included unauthorized access attempts to protected routes.

Table 4.2: Test Cases – Authentication Module

TC ID	Test Description	Input	Expected Output	Status
TC-01	Valid user registration	Valid email + strong password	Account created, redirect to login	Pass
TC-02	Duplicate email registration	Already registered email	Error: Email already exists	Pass
TC-03	Valid user login	Correct email and password	JWT issued, redirect to dashboard	Pass

TC-04	Invalid login credentials	Wrong password	Error: Invalid credentials	Pass
TC-05	Access protected route without token	No JWT token in header	401 Unauthorized error	Pass

Table 4.3: Test Cases – Document Management Module

TC ID	Test Description	Input	Expected Output	Status
TC-06	Upload valid document	PDF file under 10 MB	Document listed in dashboard	Pass
TC-07	Upload oversized file	File exceeding 10 MB	Error: File size exceeds limit	Pass
TC-08	Download document	Valid document ID	File downloaded to device	Pass
TC-09	Delete document	Valid document ID	Document removed from DB and storage	Pass
TC-10	Search by category	Category: Medical Records	Filtered list of medical records	Pass

Table 4.4: Test Cases – Appointment Module

TC ID	Test Description	Input	Expected Output	Status
TC-11	Schedule new appointment	Valid date, time, type	Appointment saved, reminder scheduled	Pass
TC-12	Schedule past appointment	Past date and time	Error: Cannot schedule past date	Pass

TC-13	Edit appointment	Updated date/time	Record updated, reminder rescheduled	Pass
TC-14	Cancel appointment	Valid appointment ID	Record removed, reminder cancelled	Pass
TC-15	Receive email reminder	Appointment time approaching	Email reminder sent to user	Pass

4.3 Installation Instructions

Prerequisites

- Node.js v16+ and npm must be installed.
- MongoDB (local) or MongoDB Atlas account.
- Git for cloning the repository.

Backend Installation

10. Clone repository: `git clone <repository-url>`
11. Navigate to backend: `cd backend`
12. Install dependencies: `npm install`
13. Create `.env` file with: `PORT`, `MONGO_URI`, `JWT_SECRET`, `EMAIL_USER`, `EMAIL_PASS`.
14. Start server: `npm start` (or `npm run dev` for development with nodemon).

Frontend Installation

15. Navigate to frontend: `cd frontend`
16. Install dependencies: `npm install`
17. Create `.env` with: `REACT_APP_API_URL=http://localhost:5000`
18. Start app: `npm start` — opens at `http://localhost:3000`

4.4 End User Instructions

Registration and Login

19. Open the application URL in a web browser.
20. Click 'Register' and fill in name, email, password, and account type.
21. Login with registered credentials to access the dashboard.

Managing Documents

22. Navigate to 'Documents' from the sidebar.
23. Click 'Upload Document', select the file, choose category, and add optional tags.
24. Click 'Upload' to save. Download or delete using the action icons.
25. Use the search bar and filter dropdown for quick retrieval.

Managing Appointments

26. Navigate to 'Appointments' from the sidebar.
27. Click '+ Schedule' to add an appointment with title, type, date, time, and reminder.
28. Reschedule or cancel appointments using the action buttons on appointment cards.

Managing Medications

29. Navigate to 'Medications' from the sidebar.
30. Click '+ Add Medication', enter drug name, dosage, frequency, and refill date.
31. Use 'Request Refill', 'Details', or 'Remove' buttons as needed.

CHAPTER 5 RESULTS AND DISCUSSIONS

5.1 Module Description

Authentication Module

The Authentication Module handles all user registration, login, and session management operations. Users create accounts by providing name, email address, and password. The system validates input data, hashes passwords using bcrypt before storage, and issues JWT tokens upon successful login. The module enforces session security by requiring valid tokens for all protected API routes and automatically invalidating tokens upon logout or expiry.

Profile Management Module

The Profile Management Module enables users to create and manage individual profiles within the system. Parent users can view and edit their own profile details and create multiple child profiles linked to their account. Each profile stores personal information including name, date of birth, contact details, and an optional profile photo. The module enforces role-based access so that child profiles can only be accessed within the child user's scope.

Document Management Module

The Document Management Module is the core functional component of the system. It allows authenticated users to upload health documents in supported formats (PDF, JPEG, PNG). Uploaded files are stored in cloud storage and their metadata is saved in MongoDB. Users can view, download, and delete their documents. Documents are organized by predefined categories: Medical Records, Prescriptions, Vaccination Records, Diagnostic Reports, School Documents, Aadhaar, and Birth Certificate. A search function allows users to quickly locate documents by name, category, or tag.

Appointment Management and Reminder Module

The Appointment Management Module allows users to schedule health appointments for any of their linked profiles. When creating an appointment, users specify the title, type (in-person or virtual), date, time, location or telehealth link, and reminder settings. The system supports calendar view and list view of appointments. The reminder system operates through scheduled background jobs that check appointment dates and trigger email notifications using NodeMailer. Appointments can be rescheduled or cancelled at any time.

Medication Tracking Module

The Medication Tracking Module enables users to maintain an organized record of current medications for each profile. For each medication entry, users can record the drug name, dosage, frequency (e.g., once daily), and refill date. The module supports refill request actions and detailed medication views. This feature is particularly valuable for managing medications for children or elderly family members, ensuring nothing is missed in their health regimes.

Notification Module

The Notification Module manages the delivery and display of system notifications. These include appointment reminders sent via email, system alerts for successful document uploads, and confirmation notifications for scheduled appointments. Notifications are displayed on the user dashboard as a notification bell icon with a badge count. Users can view and dismiss individual notifications, and the system maintains a log of recent activity on the dashboard.

5.2 User Interface Snapshots

The following section presents the actual user interface of the Centralized Web-Based Health Document and Appointment Management System (HealthVault – Document Manager) with detailed descriptions of each screen.

5.2.1 Dashboard Screen

The dashboard is the main landing screen after successful login. It presents a personalized welcome message for the logged-in user profile (shown as 'Welcome back, Lily'). The top section displays three summary widgets: Total Documents (8, with +3 this month), Upcoming Appointments (2, with the next on Mar 10), and Action Needed (1, Review new results). Below the widgets, the Health Documents section lists recently uploaded files with their category, provider, date, and file size. A search bar and filter button enable quick document retrieval. The right panel shows an Appointments calendar in month view with today's date highlighted. The bottom section shows Current Medications (Cetirizine 5mg, once daily) and a Recent Activity feed showing timeline events. The navigation sidebar provides access to Dashboard, Documents, Appointments, and Medications modules.

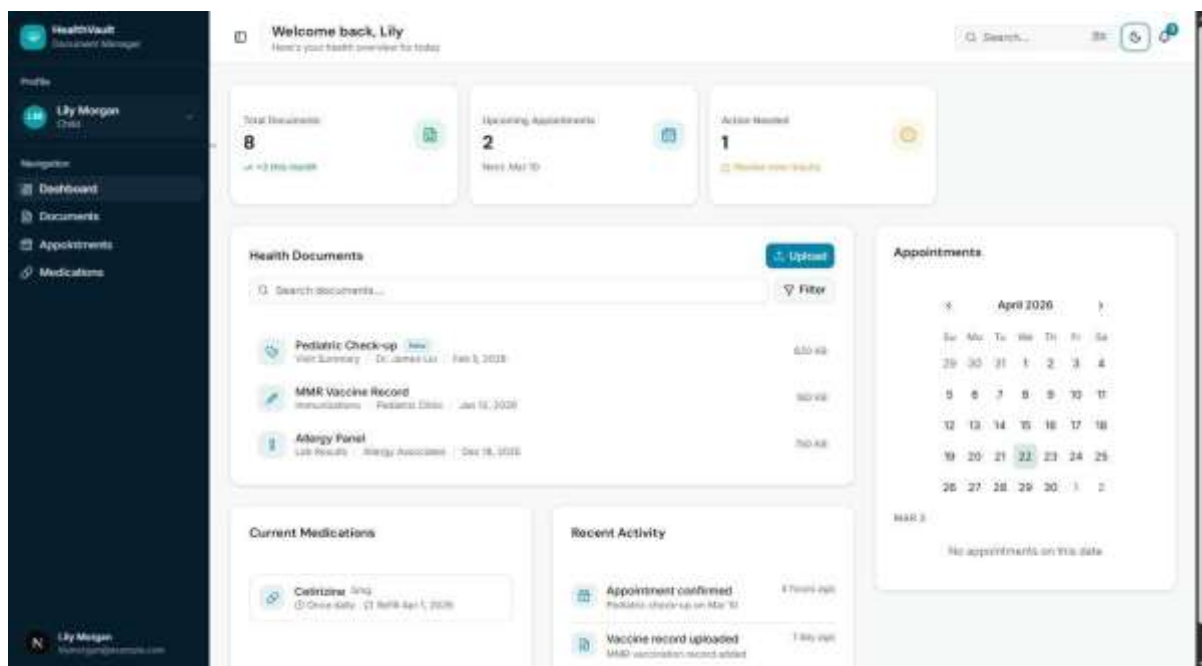


Figure 5.1: Dashboard Screen – HealthVault Document Manager

5.2.2 Health Documents Screen

The Documents screen presents a focused view of the health documents for the currently selected profile (Viewing documents for Lily). A full-width search bar and filter button are displayed at the top for efficient document retrieval. Each document is listed as a row showing a category icon, document name, category label (e.g., Visit Summary, Immunizations, Lab Results), provider or facility name, date of upload, and file size. The 'New' badge highlights recently uploaded documents. An 'Upload' button in the top right opens the document upload workflow. The three documents shown are Pediatric Check-up (620 KB), MMR Vaccine Record (180 KB), and Allergy Panel (750 KB), demonstrating a realistic distribution of pediatric health records.

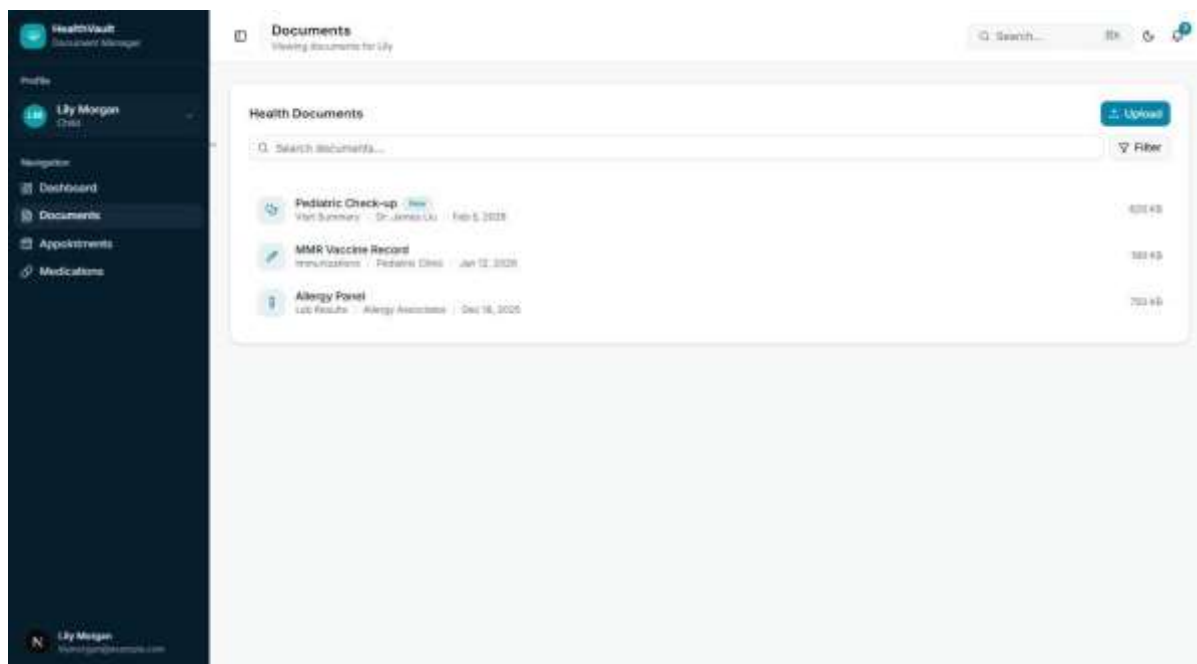


Figure 5.2: Health Documents Screen

5.2.3 Appointments Screen

The Appointments screen (Viewing appointments for Lily) shows a split layout with a calendar on the left and an appointment list on the right. The calendar shows April 2026 with the current date (22nd) highlighted in teal. The 'ALL UPCOMING' section on the right lists two upcoming appointments as cards. The first is 'Pediatric Check-up' (in-person badge) with Dr. James Liu on Mar 10 at 11:30 AM at Pediatric Clinic, with Reschedule and Cancel action buttons. The second is 'Allergy Follow-up' (virtual badge) with Dr. Aisha Patel on Mar 20 at 2:00 PM via Telehealth. The '+ Schedule' button at the top right allows users to add new appointments. The dual-badge system (in-person vs virtual) clearly differentiates appointment types.

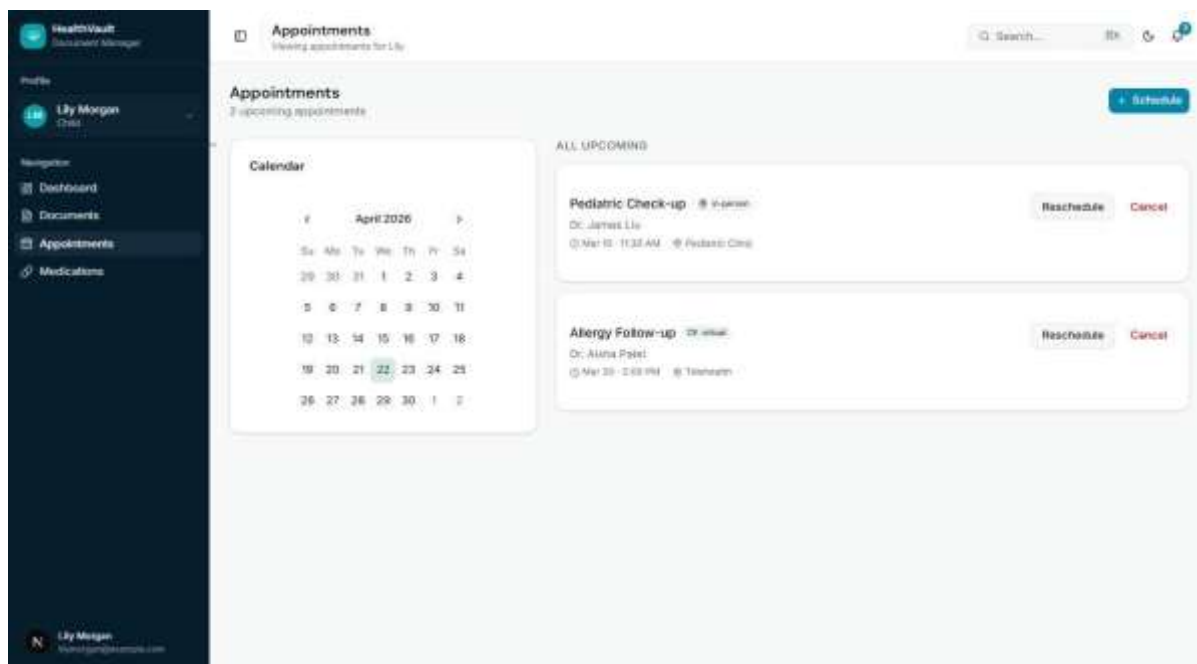


Figure 5.3: Appointments Screen

5.2.4 Medications Screen

The Medications screen (Viewing medications for Lily) shows a clean, card-based layout listing active medications. Currently showing 1 active medication, the card displays Cetirizine 5mg with a dosage icon, frequency (Once daily), and refill reminder (Refill by Apr 1, 2026). Three action buttons are provided for each medication card: 'Request Refill' for initiating a refill request, 'Details' for viewing the full medication information, and 'Remove' (in red) for discontinuing the medication. The '+ Add Medication' button at the top right opens the add medication workflow. The card-based design ensures medications are easily readable and accessible.

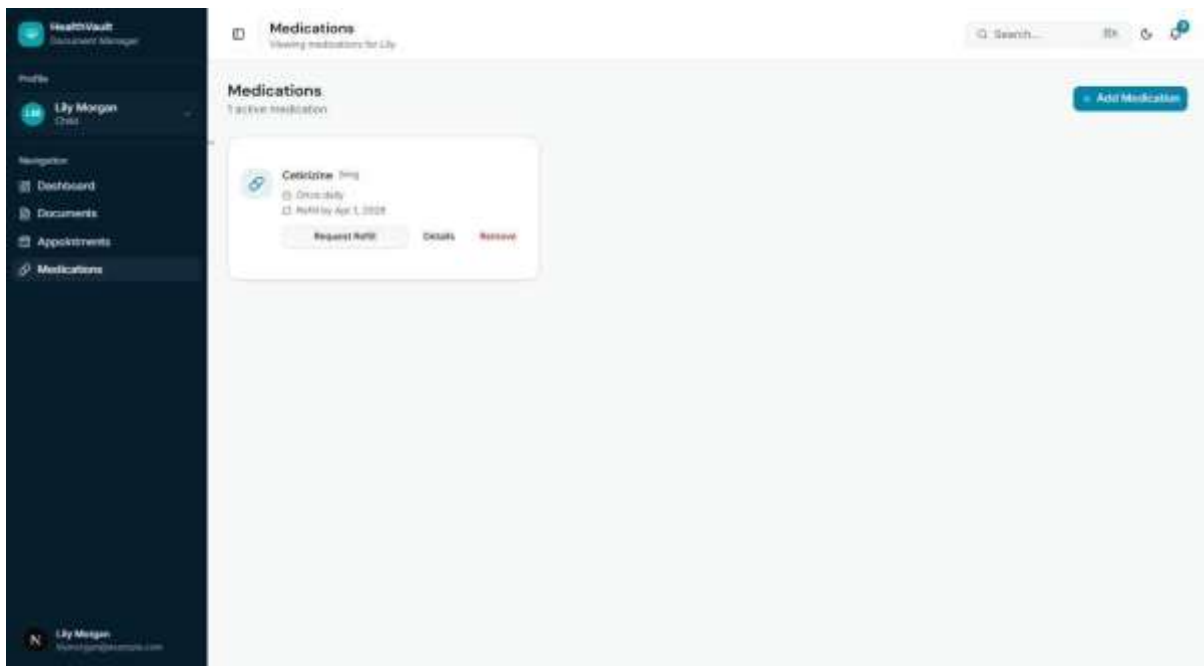


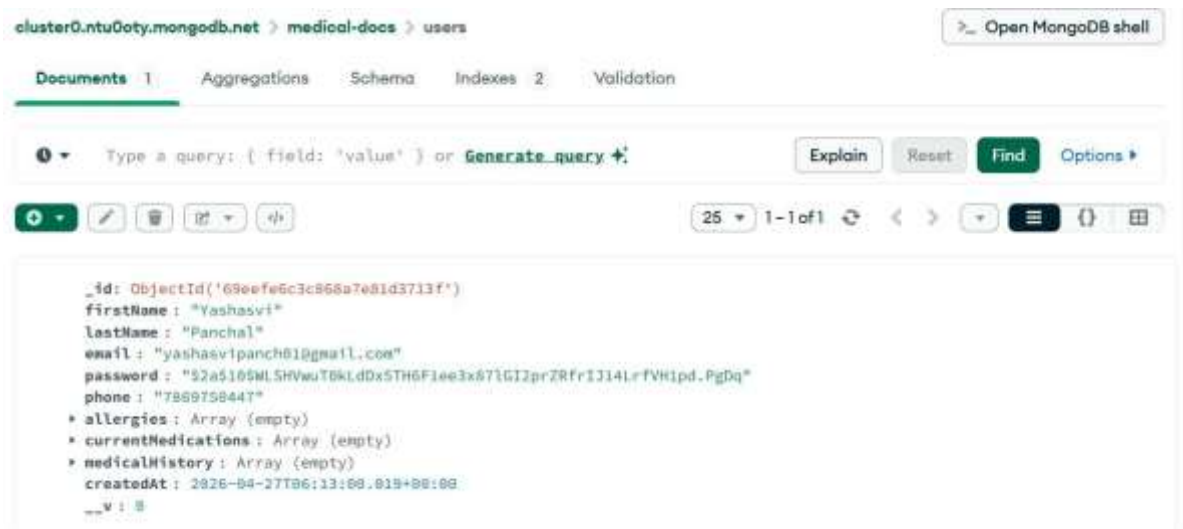
Figure 5.4: Medications Screen

5.3 Database Tables

The following tables describe the database collection field structures as implemented in the MongoDB database:

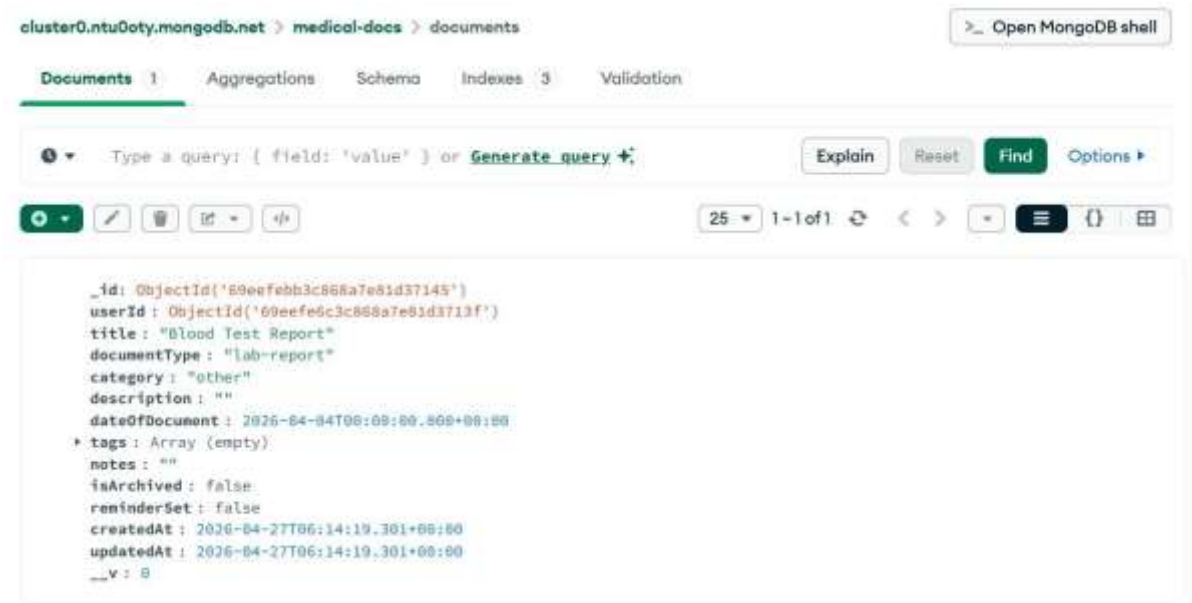


Table 5.1: User Collection Fields



Field Name	Data Type	Constraints	Description
_id	ObjectId	Primary Key	Unique user identifier
email	String	Required,Unique	Email address for login
passwordHash	String	Required	bcrypt hashed password string
accountType	String(Enum)	Required	'parent' or 'child'
createdAt	Date	Auto-generated	Account creation timestamp
lastLogin	Date	Optional	Timestamp of last login

Table 5.2: Document Collection Fields



Field Name	Data Type	Constraints	Description
_id	ObjectId	Primary Key	Unique document identifier
profileId	ObjectId(Ref)	Required,FK	Reference to associated profile
categoryId	ObjectId(Ref)	Required,FK	Reference to document category
fileName	String	Required	Original name of uploaded file
filePath	String	Required	Cloud storage URL
fileType	String	Required	MIME type of document
fileSize	Number	Required	File size in bytes

uploadDate	Date	Auto-gen	Upload timestamp
tags	Array[String]	Optional	User-defined tags

Table 5.3: Appointment Collection Fields

Field Name	Data Type	Constraints	Description
_id	ObjectId	Primary Key	Unique appointment identifier
profileId	ObjectId(Ref)	Required, FK	Reference to associated profile
title	String	Required	Appointment title
appointmentType	String	Required	in-person or virtual
description	String	Optional	Additional appointment details
appointmentDate	Date	Required	Scheduled date and time
location	String	Optional	Physical location or telehealth link
reminderDateTime	Date	Optional	Date/time to trigger reminder
status	String(Enum)	Default:Upcoming	Upcoming, Completed, Cancelled

CHAPTER 6 SUMMARY AND CONCLUSIONS

6.1 Project Summary

The Centralized Web-Based Health Document and Appointment Management System was designed and developed to address the growing need for an organized, secure, and accessible digital platform for managing personal health records, appointments, and medications for individuals and families.

The project involved a comprehensive development process beginning with a thorough analysis of requirements derived from a review of existing healthcare platforms and academic literature, followed by system design, implementation, and testing. The system was built using a modern full-stack JavaScript technology stack comprising React.js for the frontend, Node.js and Express.js for the backend, and MongoDB for the database.

The system successfully implements the following core functionalities:

- User authentication and authorization using email/password login with JWT-based session management and bcrypt password security.
- Multi-profile management supporting parent and child user accounts with role-based access control ensuring appropriate data segregation.
- Comprehensive document management including upload, categorization, search, view, download, and deletion of health documents with cloud storage integration.
- Appointment scheduling and management with automated email reminder notifications, calendar view, and support for both in-person and virtual appointment types.
- Medication tracking module for recording current medications, dosage schedules, and refill reminders.
- A notification system providing real-time dashboard alerts and email reminders for upcoming appointments.
- A clean, responsive user interface providing an intuitive experience accessible on desktop, tablet, and mobile devices.

The literature review conducted at the beginning of this project confirmed that the majority of existing healthcare management platforms are either individual-centric and lack family profile management, or provide document management and appointment scheduling as separate, disconnected tools. The proposed system bridges this gap by integrating all these functionalities into a single unified platform.

The user interface, as evidenced by the application screenshots, achieves the desired balance of functionality and simplicity. The HealthVault interface uses a teal color scheme with a dark sidebar, providing clear visual hierarchy and intuitive navigation. The dashboard provides an immediate health overview through summary widgets, while individual module screens maintain focused, clutter-free layouts.

6.2 Conclusions

Based on the design, development, and testing of the system, the following conclusions are drawn:

First, the system successfully achieves its primary objective of providing a centralized digital repository for health documents. By enabling users to upload, categorize, and retrieve medical records in a structured manner, the system eliminates reliance on paper-based record keeping and reduces the risk of document loss or misplacement.

Second, the multi-profile management feature addresses a significant limitation in existing healthcare platforms. By enabling a single parent account to manage health records for multiple children, the system provides a genuinely family-centric solution that supports the diverse needs of modern families.

Third, the appointment reminder system proved effective in helping users maintain awareness of upcoming health appointments. Automated email notifications reduce the cognitive burden on users and have the potential to significantly improve appointment adherence, contributing to better health outcomes.

Fourth, the medication tracking module addresses a critical healthcare challenge by providing an organized, easily accessible record of medications with refill reminders. This feature directly supports medication adherence and reduces the risk of missed doses.

Fifth, the implementation of security measures including bcrypt password hashing, JWT authentication, role-based access control, and HTTPS communication ensures that sensitive health data is adequately protected from unauthorized access.

Sixth, the system's responsive design ensures accessibility across a wide range of devices, making health record management convenient and available to users regardless of their preferred device.

In conclusion, the Centralized Web-Based Health Document and Appointment Management System represents a meaningful contribution toward the digital transformation of personal healthcare management. The system demonstrates that integrating document management, appointment scheduling, medication tracking, and family profile management into a single web-based platform is both technically feasible and practically valuable.

CHAPTER 7 FUTURE SCOPE

7.1 Potential Enhancements

The system offers significant potential for future expansion. The following areas represent the most promising directions for future development:

Mobile Application Development

Dedicated native mobile applications for Android and iOS platforms would significantly enhance the user experience. Mobile applications can leverage push notifications for appointment reminders, offline document access, camera integration for scanning physical documents, and biometric authentication for enhanced security. A Progressive Web App (PWA) version could be developed as an intermediate step, providing mobile-app-like functionality through the browser.

Telemedicine Integration

A future version could integrate with telemedicine platforms to enable video consultations between patients and healthcare providers. During such consultations, authorized doctors could access the patient's medical history and uploaded documents directly through the platform. The existing appointment system's virtual appointment type provides a natural foundation for this integration.

Artificial Intelligence and Machine Learning

The integration of AI and machine learning presents exciting opportunities for system enhancement. Potential AI-driven features include automated document categorization based on content analysis using optical character recognition (OCR), intelligent health insights and personalized recommendations based on medical history, predictive reminders for preventive care such as vaccination schedules and annual check-ups, and natural language processing for searching documents by content rather than just metadata.

Integration with Healthcare Providers

Future versions could support direct integration with hospital management systems, laboratories, and diagnostic centres. This would enable automatic import of medical reports and test results into a patient's digital health record, eliminating the need for manual uploads. Standards such as HL7 FHIR (Fast Healthcare Interoperability Resources) could be implemented to ensure seamless data exchange with electronic health record systems.

Blockchain-Based Data Security

Blockchain technology could be explored for immutable record-keeping in future versions. Blockchain-based audit trails would provide tamper-proof logs of all document access and modifications, giving patients greater transparency and control over who has accessed their health information. Smart contracts could automate consent management for data sharing with healthcare providers.

Wearable Device Integration

Integration with wearable health monitoring devices such as smartwatches and fitness trackers could enable automatic collection and storage of health metrics including heart rate, blood pressure, step count, sleep patterns, and blood glucose levels. This would transform the platform from a passive document storage system into an active, continuous health monitoring tool.

Advanced Analytics Dashboard

A future analytics module could provide users with visual summaries of their health history, appointment frequency trends, medication adherence patterns, and document upload statistics.

Graphs and charts showing health trends over time would help users and their healthcare providers identify patterns and make more informed health decisions. Healthcare providers with appropriate authorization could access anonymized aggregate data for population health research.

Multi-Language Support

To improve accessibility for users in India's diverse linguistic landscape, the system could be extended to support multiple regional languages through internationalization (i18n) implementation. This would enable a significantly broader user base to effectively use the platform in their preferred language, particularly supporting users in rural and semi-urban areas.

Online Payment Integration

Future versions could incorporate online payment functionality to enable users to pay for medical appointment fees, laboratory test bookings, or prescription orders directly through the platform. Integration with popular Indian payment gateways such as Razorpay or PayU would make the platform a comprehensive one-stop solution for healthcare management.

National Health Digital Infrastructure Alignment

The system could be developed to align with India's Ayushman Bharat Digital Mission (ABDM) by integrating with the Unified Health Interface (UHI) and supporting Health ID (ABHA) creation. This would enable users to connect their health records with the national digital health ecosystem, improving interoperability with government and private healthcare providers across India.

Emergency Access Feature

A critical enhancement would be the implementation of an emergency access feature allowing users to create a QR code or emergency access token that contains essential health information. In emergency situations, first responders or emergency room staff could scan this code to instantly access critical medical information such as blood type, allergies, current medications, and emergency contacts, potentially being life-saving.

APPENDIX

Appendix A: API Endpoint Reference

The following table provides a reference summary of the main RESTful API endpoints implemented in the backend:

Method	Endpoint	Auth	Description
POST	/api/auth/register	No	Register a new user account
POST	/api/auth/login	No	Login and receive JWT token
GET	/api/profiles	Yes	Get all profiles for authenticated user
POST	/api/profiles	Yes	Create a new profile
PUT	/api/profiles/:id	Yes	Update profile by ID
DELETE	/api/profiles/:id	Yes	Delete profile by ID
GET	/api/documents	Yes	Get all documents for a profile
POST	/api/documents/upload	Yes	Upload a new document
DELETE	/api/documents/:id	Yes	Delete document by ID
GET	/api/appointments	Yes	Get all appointments for a profile
POST	/api/appointments	Yes	Create a new appointment
PUT	/api/appointments/:id	Yes	Update appointment by ID
DELETE	/api/appointments/:id	Yes	Delete appointment by ID

GET	/api/medications	Yes	Get all medications for a profile
POST	/api/medications	Yes	Add a new medication
PUT	/api/medications/:id	Yes	Update medication by ID
Method	Endpoint	Auth	Description
DELETE	/api/medications/:id	Yes	Remove medication by ID

BIBLIOGRAPHY

References

- [1] Tang, P. C., et al., "Personal Health Records: Definitions, Benefits, and Strategies for Overcoming Barriers to Adoption," *Journal of the American Medical Informatics Association*, 2006.
- [2] Roehrs, A., et al., "Personal Health Records: A Systematic Literature Review," *Journal of Medical Internet Research*, vol. 19, no. 1, e13, 2017.
- [3] Alanazi, A., et al., "Personal Health Record (PHR) Experience and Recommendations for a Transformation in Saudi Arabia," *Journal of Personalized Medicine*, 2023.
- [4] Gabbay, J. M., et al., "An Approach to Family-Centered Electronic Health Records in Pediatrics," *Journal of Patient Experience*, 2022.
- [5] Blumenthal, D., & Tavenner, M., "The 'Meaningful Use' Regulation for Electronic Health Records," *New England Journal of Medicine*, vol. 363, no. 6, pp. 501-504, 2010.
- [6] Jha, A. K., et al., "Use of Electronic Health Records in U.S. Hospitals," *New England Journal of Medicine*, vol. 360, no. 16, pp. 1628-1638, 2009.
- [7] Kellermann, A. L., & Jones, S. S., "What It Will Take to Achieve the As-Yet-Unfulfilled Promises of Health Information Technology," *Health Affairs*, vol. 32, no. 1, pp. 63-68, 2013.
- [8] Adler-Milstein, J., & Jha, A. K., "HHS Efforts to Improve Interoperability and Patient Access to Health Data," *Health Affairs*, vol. 36, no. 4, pp. 637-641, 2017.
- [9] Mandel, J. C., et al., "SMART on FHIR: A Standards-Based, Interoperable Apps Platform for Electronic Health Records," *Journal of the American Medical Informatics Association*, vol. 23, no. 5, pp. 899-908, 2016.
- [10] Roehrs, A., et al., "Personal Health Records: A Systematic Literature Review," *JMIR*, 2017.
- [11] Patel, V., et al., "A Web-Based Medical Appointment System for Hospitals," *International Journal of Scientific & Technology Research*, 2020.
- [12] Gupta, A., et al., "Design and Development of a Web-Based Hospital Management System," *International Journal of Computer Applications*, 2019.
- [13] Singh, D., & Kaur, G., "Integrated e-Healthcare Management System Using Web Technologies," *International Journal of Information Management Data Insights*, 2024.
- [14] Raghupathi, W., & Raghupathi, V., "Big Data Analytics in Healthcare: Promise and Potential," *Health Information Science and Systems*, vol. 2, no. 3, 2014.
- [15] Ozair, F., et al., "Ethical Issues in Electronic Health Records: A General Overview," *Perspectives in Clinical Research*, vol. 6, no. 2, pp. 73-76, 2015.
- [16] Pressman, R. S., & Maxim, B. R., *Software Engineering: A Practitioner's Approach*, 9th Edition. McGraw-Hill, 2019.
- [17] Sommerville, I., *Software Engineering*, 10th Edition. Pearson Education, 2016.
- [18] Laudon, K. C., & Laudon, J. P., *Management Information Systems*. Pearson Education, 2020.
- [19] World Health Organization, *Digital Health Systems and Global Healthcare Development*. WHO Press, 2021.
- [20] Ministry of Health and Family Welfare, Government of India, *Digital Health Initiatives and National Health Mission Reports*, 2022.

